

GEP Intern – Technology: 5-Day Sprint Plan

Execute only — topics + verified links (click or copy URL)

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Anchors: GEP_Prep_Roadmap.pdf + GEP_Question_Practice_Roadmap.pdf | ~6–7 hrs/day | Day 1 = DBMS rebuild

How to use

- Each day: **Topics** → **Read** → **Practice**. Tick and move on.
- Every resource shows the **full URL** under the title — if a PDF click fails, copy-paste into browser.
- Old GFG /quizzes/...-gq/ links are dead (Quiz Hub). Not used here.

Day 1 — AI (light) + DBMS depth rebuild + OOP

Block A — AI MCQs (~2 h)

Topics: AI vs ML vs DL; supervised / unsupervised; LR *idea*; CV / RL use-cases; NLP 1-liner.

Read:

- **AI vs ML vs DL vs NN**
<https://www.geeksforgeeks.org/machine-learning/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks/>
- **Intro to Machine Learning**
<https://www.geeksforgeeks.org/machine-learning/introduction-machine-learning/>
- **Linear Regression**
<https://www.geeksforgeeks.org/machine-learning/ml-linear-regression/>
- **Computer Vision**
<https://www.geeksforgeeks.org/computer-vision/computer-vision/>
- **Reinforcement Learning**
<https://www.geeksforgeeks.org/machine-learning/what-is-reinforcement-learning/>

Practice:

- ☐ **AI aptitude — 25 MCQs w/ answers**
<https://www.javaguides.net/2024/05/ai-aptitude-test-25-mcq-questions.html>

Block B — DBMS depth rebuild (~3.5–4 h) — DO NOT SKIP

Topics (in order): DBMS vs files; keys (super/candidate/PK/FK); ER; SQL DDL/DML; INNER/LEFT/RIGHT/FULL joins; GROUP BY/HAVING; 1NF–3NF; indexing idea; ACID; concurrency names.

Read (this order):

- **Intro to DBMS**
<https://www.geeksforgeeks.org/dbms/introduction-of-dbms-database-management-system-set-1/>
- **Types of Keys**
<https://www.geeksforgeeks.org/dbms/types-of-keys-in-relational-model-candidate-super-primary-alternate-and-foreign/>
- **ER Model**
<https://www.geeksforgeeks.org/dbms/introduction-of-er-model/>
- **SQL command types (DDL/DML/...)**
<https://www.geeksforgeeks.org/sql/sql-ddl-dql-dml-dcl-tcl-commands/>
- **SQL Joins**
<https://www.geeksforgeeks.org/sql/sql-join-set-1-inner-left-right-and-full-joins/>
- **SQL GROUP BY**
<https://www.geeksforgeeks.org/sql/sql-group-by/>
- **Normal Forms**
<https://www.geeksforgeeks.org/dbms/normal-forms-in-dbms/>
- **Indexing**
<https://www.geeksforgeeks.org/dbms/indexing-in-databases-set-1/>
- **ACID**
<https://www.geeksforgeeks.org/dbms/acid-properties-in-dbms/>
- **Concurrency (skim)**
<https://www.geeksforgeeks.org/dbms/concurrency-control-in-dbms/>
- **DBMS Tutorial hub (if lost)**
<https://www.geeksforgeeks.org/dbms/dbms/>

Practice:

- ☐ **IndiaBIX — Database MCQs**
<https://www.indiabix.com/database/questions-and-answers/>
- ☐ **GFG — DBMS Interview Qs**
<https://www.geeksforgeeks.org/commonly-asked-dbms-interview-questions/>
- ☐ **GFG quiz — Normal Forms**
<https://www.geeksforgeeks.org/quizzes/normal-forms/>
- ☐ **W3Schools — SQL Quiz**
https://www.w3schools.com/sql/sql_quiz.asp
- ☐ By hand: 2 INNER + 2 LEFT JOIN queries; speak “2NF vs 3NF” and “ACID” aloud

Block C — OOP (~30–40 min)

- ☐ **GFG — OOP Concepts quiz**
<https://www.geeksforgeeks.org/quizzes/oop-concepts-gq/>
- ☐ **GFG — OOP Interview Qs**
<https://www.geeksforgeeks.org/interview-prep/oops-interview-questions/>
- Speak 4 pillars <2 min

Done bar: AI once | DBMS reads ticked | IndiaBIX + interview Qs + SQL quiz | 4 JOINS by hand | OOP spoken

Day 2 — Named algos + Easy coding + JS

Topics

Kadane; Binary Search; BFS/DFS; hashing / Two Sum; strings; JS `==/===`, `map/filter`. Speak 30s: Kadane, BS, BFS/DFS, Dijkstra idea, Merge/Quick, hashing, prefix sums.

Read (only if stuck)

- **Kadane explained**

<https://www.geeksforgeeks.org/dsa/largest-sum-contiguous-subarray/>

- **GEP Summer Intern 2026 write-up**

<https://leetcode.com/discuss/post/7161376/gep-worldwide-summer-internship-intervie-dvv5/>

Practice (code + dry-run aloud)

- ☐ **LC — Maximum Subarray**

<https://leetcode.com/problems/maximum-subarray/>

- ☐ **LC — Binary Search**

<https://leetcode.com/problems/binary-search/>

- ☐ **LC — Level Order (BFS)**

<https://leetcode.com/problems/binary-tree-level-order-traversal/>

- ☐ **LC — Two Sum**

<https://leetcode.com/problems/two-sum/>

- ☐ **LC — First Unique Character**

<https://leetcode.com/problems/first-unique-character-in-a-string/>

- ☐ **LC — Valid Parentheses**

<https://leetcode.com/problems/valid-parentheses/>

- ☐ **W3Schools — JS Quiz**

https://www.w3schools.com/js/js_quiz.asp

Done bar: Kadane + BS + BFS coded | ≥ 3 more Easy | JS quiz | named list spoken

Day 3 — OS + SQL drills + Projects + more Easy

Block A — OS (~1.5–2 h)

Topics: process vs thread; scheduling names; deadlock (4 conditions); paging idea; mutex/semaphore 1-liner.

Read:

- **OS Intro**

<https://www.geeksforgeeks.org/operating-systems/introduction-of-operating-system-set-1/>

- **Process Management**

<https://www.geeksforgeeks.org/operating-systems/introduction-of-process-management/>

- **Deadlock**

<https://www.geeksforgeeks.org/operating-systems/introduction-of-deadlock-in-operating-system/>

- **Paging**

<https://www.geeksforgeeks.org/operating-systems/paging-in-operating-system/>

- **OS Tutorial hub**

<https://www.geeksforgeeks.org/operating-systems/operating-systems/>

Practice:

- ☐ **Examveda — OS MCQs**

<https://www.examveda.com/computer-fundamentals/practice-mcq-question-on-operating-system/>

- ☐ **GFG — OS Interview Qs**

<https://www.geeksforgeeks.org/operating-systems/operating-systems-interview-questions/>

Block B — SQL hand drills (DBMS retention) (~1 h)

- ☐ Write cold: INNER + LEFT JOIN; one GROUP BY+HAVING

- ☐ Re-do Day 1 IndiaBIX / SQL quiz wrongs only

- ☐ **W3Schools — SQL Exercises**

https://www.w3schools.com/sql/sql_exercises.asp

Block C — Projects + Easy

- ☐ Speak: IFFCO, LogiFlow(+SSR), Community Hero, AirHelp — role, impact, 2–3 follow-ups

- ☐ **LC — Number of Islands**

<https://leetcode.com/problems/number-of-islands/>

- ☐ **LC — Buy/Sell Stock**

<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/>

Done bar: OS MCQs + interview Qs | JOIN cold | 4 projects deep | ≥ 5 Easy total

Day 4 — Full mock OA

Platform feel

- **How to take iMocha**

<https://help.imocha.io/kb/article/251/how-to-take-a-test-on-imocha>

Mock sitting (~70–90 min)

1. AI checklist / Day 1 wrongs — 15 min

2. DBMS flash: keys, joins, 1–3NF, ACID (speak) — 15 min

3. One Easy timed (strings/hashing) — 25 min

4. JS fill-in — 15 min

5. OS weak spots — 15 min

Done bar: one full mock; note only misses

Day 5 — Interview + HR + warm-up

Topics

R1: clarify → approach → code → dry-run. R2: projects. HR: Hyd/Mum #1. Company: AI procurement / supply chain; Quantum, SMART, NEXXE. Open to Angular/.NET. End-Q: scale / AI agents / stack.

Warm-up

- ☐ Re-solve 2 Easy from Day 2
- ☐ Kadane dry-run on `[-2,1,-3,4,-1,2,1,-5,4]`
- ☐ Speak: one algo + one project + “ACID in one minute”
- ☐ Print resume: `ojas_srivastava_gep_technology_intern.pdf`

Done bar: resume printed | location + end-Q | sleep

Final ready bar

- ☐ AI once ☐ DBMS depth + IndiaBIX/SQL quiz ☐ OS MCQs ☐ OOP quiz
- ☐ Named algos spoken ☐ Kadane + BS + BFS coded
- ☐ ≥ 5 Easy ☐ JS quiz ☐ Projects deep ☐ Mock OA

Links re-verified. Avoid GFG *-gq quiz URLs — they redirect to empty Quiz Hub.